

Physics-Informed Digital Twin Integrating Raman Distributed Fibre Optic Sensing for Predictive Maintenance of Utility-Scale Solar Farm Inverters

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ABSTRACT

Purpose of the study: Reliable operation of utility-scale photovoltaic (PV) systems are critical to the global renewable energy transition with solar energy having the largest growth of all renewables in 2025 ^[1]. Power electronic solar inverters remain among the most failure-prone components within PV plants ^[2] and are exposed to thermal-electrical degradation mechanisms that contribute to high maintenance costs and energy-yield losses that scale ^[3].

Methodology: This research proposes a physics-informed digital twin coupling a reduced-order thermal model of a string inverter with a detailed simulated Raman distributed temperature sensing (DTS) model. Spatial temperature profiles obtained from the sensing system are compared with model predictions across multiple inverter units and validated with real world data.

Key Findings: Raman DTS simulations revealed distinct spatial and temporal thermal trends with increasing inverter degradation. The resulting physics-informed features enabled separation of the degradation classes, with independent classification performance evaluated using a held-out test set and confusion matrix with high accuracy of . Experimental validation against real-world temperature measurements will be used to quantify digital-twin prediction accuracy and assess the viability of the proposed approach beyond simulation.

Novelty/Originality: The proposed framework combines physics-based thermal modelling, distributed fibre-optic temperature sensing, and data-driven fault classification to provide spatially resolved predictive maintenance for utility-scale PV inverter fleets.

1. INTRODUCTION

As the world progresses towards an era marked by the transition to more sustainable energy alternatives, photovoltaic (PV) systems play a critical role to this energy transition, with solar energy accounting for 3TW and having the largest growth in energy sources in 2025 ^[1]. However within PV plants, power electronic inverters remain among the most failure-prone components and contribute significantly to system downtime and associated energy losses ^[2].

These string inverters are widely deployed in utility-scale photovoltaic farms because their distributed architecture allows independent maximum power point tracking and modular operation. However, these devices operate under repeated thermal and electrical stresses that can accelerate component degradation.

Important failure mechanisms include semiconductor ageing, degradation of thermal interface materials, solder fatigue, capacitor ageing, and deterioration of internal heat-transfer paths ^[4]. These mechanisms can produce characteristic changes in the spatial thermal behaviour of an inverter before catastrophic failure occurs.

Existing condition-monitoring techniques are typically based on discrete temperature, voltage, and current sensors. Although such approaches can identify abnormal operating conditions, point sensors provide limited spatial information and may detect degradation only after a localised thermal anomaly has become severe ^[5], prompting the innovation of predictive maintenance of key components.

This research therefore proposes a physics-informed digital

twin integrating a reduced-order inverter thermal model with Raman distributed temperature sensing.

2. RESEARCH AIM

The aim of this research is to develop a physics-informed digital twin capable of detecting and classifying thermal degradation in utility-scale PV string inverters using spatial temperature information obtained from Raman distributed fibre-optic sensing and validating the performance against real-world data.

3. RESEARCH OBJECTIVES

The principal objectives are:

1. Develop a reduced-order electrothermal model of a utility-scale PV string inverter.
2. Develop a Raman distributed temperature sensing model capable of reconstructing spatial temperature profiles along an inverter-mounted sensing fibre.
3. Introduce representative inverter thermal degradation mechanisms into the digital twin and quantify their effect on temporal and spatial temperature behaviour.
4. Validate the electrothermal and Raman DTS models against experimentally measured or real-world operating data.
5. Compare digital-twin predictions with measured spatial temperature profiles and quantify reconstruction and model-prediction errors.
6. Extract physics-informed thermal features and develop machine-learning methods for classification of inverter

degradation states.

7. Evaluate the robustness of degradation detection across changing operating conditions, including ambient temperature and DC-link voltage.
8. Develop and assess predictive-maintenance indicators capable of identifying emerging thermal degradation before functional inverter failure.

4. PROPOSED METHODOLOGY

The proposed framework consists of three primary components: a physics-based thermal model, a distributed photonics sensing model, and an XGBoost driven degradation-classification system.

A Cauer thermal network will represent heat generation and heat transfer between semiconductor devices, heat sinks, thermal interfaces, and the surrounding environment [6]. Model parameters will describe the dominant thermal resistances and capacitances governing inverter temperature dynamics.

Raman DTS will then be represented as a distributed measurement layer, providing temperature observations along sensing fibre positioned at thermally significant inverter locations.

Differences between digital-twin predictions and distributed sensor measurements will form residual features used to identify abnormal thermal behaviour.

5. DIGITAL TWIN ARCHITECTURE

The developed digital twin couples an electrothermal inverter model with Raman distributed temperature sensing (DTS). Semiconductor losses are propagated through a reduced-order Cauer thermal network to estimate the inverter temperature, which is mapped onto the sensing fibre to obtain the spatial temperature profile $T_{DTS}(x, t)$.

Thermal degradation is introduced through controlled increases in internal thermal resistance. Agreement between the digital-twin prediction and DTS measurement can be quantified using the thermal residual

$$r(x, t) = T_{DTS}(x, t) - T_{model}(x, t).$$

Persistent changes in the magnitude or spatial distribution of $r(x, t)$ can therefore provide an indicator of abnormal thermal behaviour. Temperature rise and spatial thermal gradients are also extracted as physics-informed features for degradation classification.

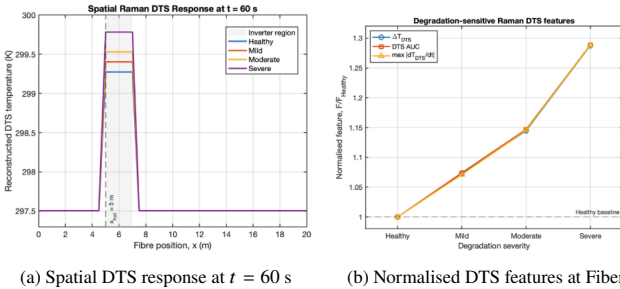


Figure 1: Raman DTS response to increasing thermal degradation: (a) spatial temperature profiles and (b) normalised thermal features relative to the healthy baseline.

6. RAMAN DISTRIBUTED TEMPERATURE SENSING

The Raman DTS model provides the spatial sensing layer of the digital twin. The predicted inverter surface temperature

is mapped onto a 20 m sensing fibre and passed through a fibre thermal-response model. The fibre is discretised at 0.5 m intervals, producing 41 spatial temperature channels.

A 1550 nm optical interrogation signal is propagated along the fibre, where temperature-dependent Stokes and anti-Stokes Raman components are generated. The anti-Stokes-to-Stokes ratio is represented by

$$R(x) = \frac{I_{AS}(x)}{I_S(x)} = C(x) \exp\left(-\frac{\Theta_R}{T(x)}\right),$$

where $C(x)$ accounts for position-dependent optical and calibration effects and

$$\Theta_R = \frac{hc\Delta\tilde{\nu}}{k_B},$$

with a Raman shift of 440 cm^{-1} [7]. Temperature is reconstructed from

$$T_{DTS}(x) = -\frac{\Theta_R}{\ln[R(x)/C(x)]}.$$

The receiver model incorporates photodetection, avalanche photodiode (APD) gain, shot and thermal noise, transimpedance amplification, and trace averaging. A 4000-trace averaging stage is implemented before temperature reconstruction [8]. Noiseless numerical testing demonstrated reconstruction of the imposed fibre-temperature profile, while differential Stokes and anti-Stokes attenuation motivates spatial calibration before final noisy-receiver evaluation.

7. XGBOOST ALGORITHM

XGBoost combines multiple decision trees to produce a final prediction, which can be expressed as

$$\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i), \quad (1)$$

where K is the number of trees and $f_k(\mathbf{x}_i)$ is the prediction from the k th tree. Features extracted from the thermal residual profiles were used as inputs to train the classifier.

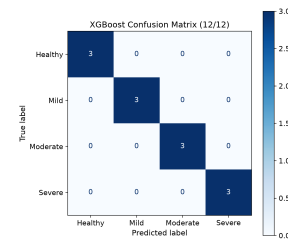


Figure 2: Confusion matrix for 12 test samples using an XGBoost model trained on 100 training points.

The trained model was evaluated using 12 previously unseen test samples. As shown in Fig. 2, all 12 samples were correctly classified, corresponding to an accuracy of

$$\text{Accuracy} = \frac{12}{12} \times 100 = 100\%. \quad (2)$$

This result indicates that the thermal residual features provide good separation between the considered operating conditions. However, the small test set limits the strength of this conclusion, and further validation using a larger and more varied dataset is required to assess the generalisation performance of the classifier as well as validation with real-world data.

8. RESULTS AND VALIDATION

The main contribution of this research is the integration of distributed fibre-optic temperature sensing with a physics-informed digital twin for inverter condition monitoring.

Unlike conventional approaches based on a small number of discrete temperature sensors, the proposed framework provides spatial information that may allow both the presence and location of thermal degradation to be identified.

This could enable maintenance to shift from reactive or schedule-based intervention toward condition-based and predictive maintenance.

9. CONCLUSION

A physics-informed digital twin integrating Raman DTS provides a promising approach for monitoring the thermal condition of string inverters across utility-scale photovoltaic farms.

By combining model predictions with spatial temperature measurements and machine-learning-based degradation classification, the proposed system aims to identify emerging inverter faults earlier and with greater spatial resolution than conventional point-sensor monitoring methods.

10. REFERENCES

1. International Energy Agency (IEA), “Global electricity demand is set to grow strongly to 2030, underscoring need for investments in grids and flexibility,” *IEA News*, 2026.
2. T. J. Formica, H. A. Khan, and M. G. Pecht, “The Effect of Inverter Failures on the Return on Investment of Solar Photovoltaic Systems,” *IEEE Access*, vol. 5, pp. 21336–21343, 2017. doi: 10.1109/ACCESS.2017.2753246.
3. X. Bao and L. Chen, “Recent Progress in Distributed Fiber Optic Sensors,” *Sensors*, vol. 12, no. 7, pp. 8601–8639, 2012. doi: 10.3390/s120708601.
4. B. Guzowski, M. Lakomski, and D. Bobinski, “Hotspot Detection in Photovoltaic Modules with Fiber Bragg Grating and Brillouin Distributed Temperature Sensors,” *Energies*, vol. 18, no. 23, Art. 6117, 2025. doi: 10.3390/en18236117.
5. [Existing reference 5: condition-monitoring / point-sensor source.]
6. [Existing reference 6: Electronics 2023 Cauer thermal-network paper.]
7. H. S. Pradhan and P. K. Sahu, “Characterisation of Raman distributed temperature sensor using deconvolution algorithms,” *IET Optoelectronics*, vol. 9, no. 2, pp. 101–107, 2015. doi: 10.1049/iet-opt.2014.0048.
8. “24 km High-Performance Raman Distributed Temperature Sensing Using Low Water Peak Fiber and Optimized Denoising Neural Network,” *Sensors*, vol. 22, no. 6, Art. 2139, 2022. doi: 10.3390/s22062139.
9. CSUC Research Data Repository, “IGBT electrical and thermal ageing data under accelerated power-cycling conditions,” Dataset, 2026. doi: 10.34810/DATA3204.

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